



FIRST TECH CHALLENGE • TEAM 19645 INFINITY

FTC Home

One place, all season. The software our team built to run a robotics season — schedule, roster, budget, build log, sponsors and live event data — designed to keep working in a gym with no signal.

FTC Home

TEAM 19645

Today

Calendar

Chat

Build

Live

Season Management

Weekly

Roster

Budget & sponsors

Plan

Parts

Scout

Archive

App

How this works

States & sync

Settings

Aarush Kikkuru

STUDENT

You are running 19645 on your own. Add a coach or mentor and this hands over automatically — you stay a student.

Add a coach

19645 • INFINITY

Aarush Kikkuru • STUDENT

Search 🔍

LIVE

MON • AUG 17 • WEEK 1

Today

GETTING STARTED • 4 OF 5

19645 Infinity is linked, so your competitions and results are already here. The rest is what only your team knows.

☒ Add the rest of the team

☒ Put your build sessions on the calendar

☒ Assign the first task

☒ Set a fundraising goal

☐ Add a build photo

One per build day is enough — the weekly page and the notebook both pull from it.

Do later

Add a build photo

How this works

NEXT COMPETITION

Illinois Qualifier — Naperville

SUN • AUG 30 • Naperville North HS

13 DAYS

SEASON OPR

—

ENGR. NB

Current

GOING

0/8

no matches yet

TODAY'S MEETING

Nothing scheduled today.

ASSIGNED TO YOU

☐ Re-cut intake side plates (2mm thinner)

Mechanical • Devin Okafor

Aug 21

IN PROGRESS

☐ Tune AprilTag alignment for far shot

Software • Aarush Kikkuru

Aug 22

IN PROGRESS

☐ Rewire odometry pods — strain relief

Electrical • Maya Rodriguez

Aug 19

T0 D0

☐ Engineering notebook — week 6 write-up

Notebook • Priya Sharma

today

T0 D0

APPROVALS

Nothing waiting on a decision.

BUDGET

\$1,455

0 open requests

left of \$5,300

BUILT BY

TEAM
19645

SCREENS

NINETEEN

RUNS ON

PHONE • LAPTOP • PIT
DISPLAY

PREPARED FOR

PROSPECTIVE
SPONSORS

WHY WE BUILT IT

A season does not fit in a group chat

An FTC team is a small organisation with a hard deadline. Ten to twenty students, a couple of volunteer mentors, a build schedule, a budget that belongs to somebody else's money, and a competition calendar nobody controls.

Most teams run that on a pile of spreadsheets, a shared drive, three group chats and one student's laptop. It works until the student graduates. Every season the same institutional knowledge — which supplier, which deadline, what we tried last year and why it failed — leaves the team wearing a graduation gown.

The tools that exist are built for offices. They assume reliable wifi, one organisation, adult users, and a subscription. A competition venue has none of those. The field runs on 5 GHz, the pits are a hostile radio environment, and the person who needs the match schedule is a sixteen-year-old holding a phone with one bar and a dying battery.

So we set three rules before writing anything

It works with no signal

Every screen reads from a database on the device. Nothing waits on the network. Sync is something that happens later, not something you stand around for.

It never invents a fact

Team identity, competitions, venues, match results and rankings come from a public API. The app stores only what no API knows — and starts genuinely empty.

Students are minors

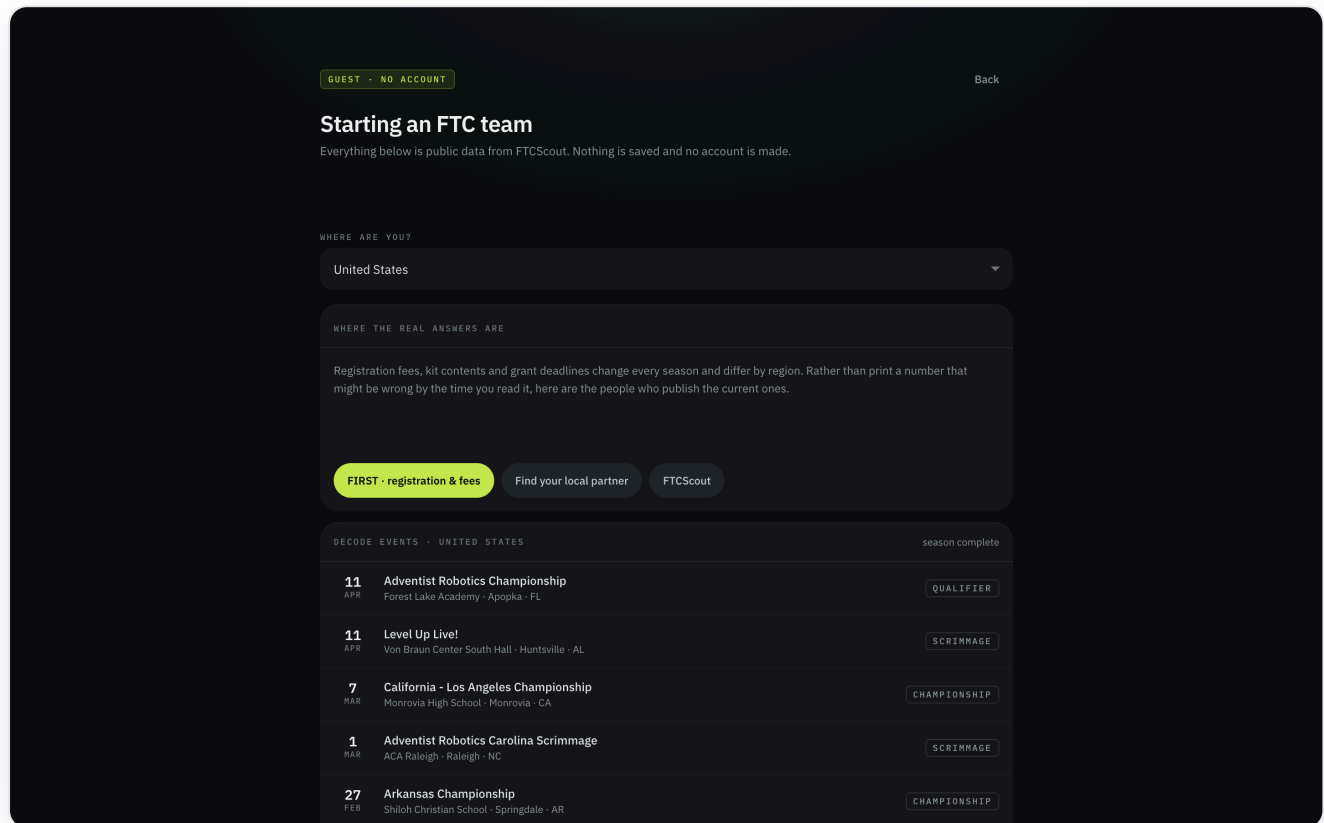
Contact details and medical records are separated from everything else and gated by role, in the database rather than in the interface.

Note on the screenshots in this document. Team 19645's identity, location and competition data are real and pulled live from FTCScout. The roster names, sponsor names, dollar figures and tasks are **sample data** created to demonstrate populated screens.

OPEN THE DOOR FIRST

Useful before you have an account

A rookie coach deciding whether to start a team should not hit a sign-up wall. The registration guide and every piece of public competition data are available with no account at all.



Guest hub. Region is pre-selected from the browser's own timezone — no permission prompt, no network call, no location data collected. Where a timezone maps to exactly one region it says so; where it covers twenty states it falls back to the country rather than assert a state it cannot know. Every event, date and venue below it is live from FTCScout.

We link out instead of guessing

Registration fees, kit contents and grant deadlines change every season and differ by region. Rather than print a number that may be wrong by the time it is read, the hub points at the organisations that publish the current ones.

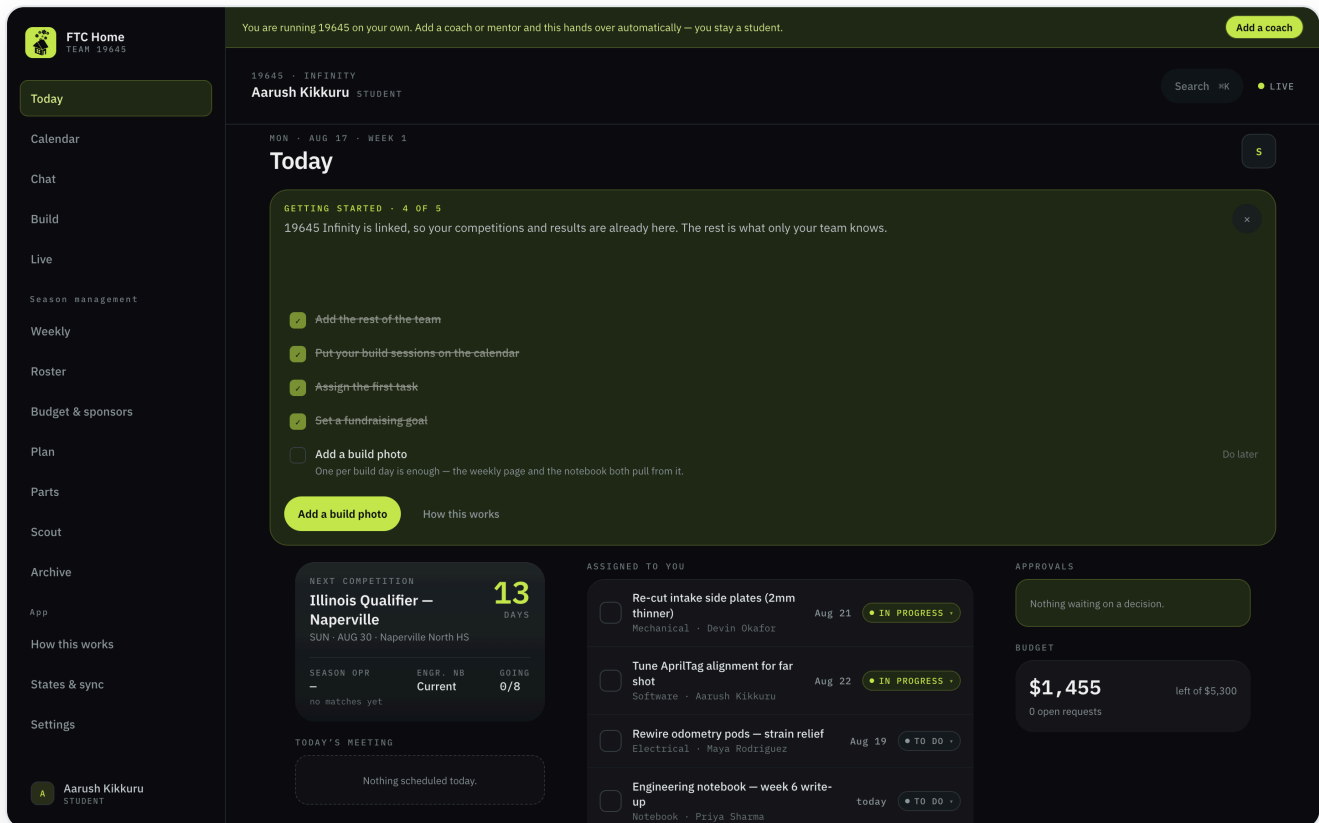
Joining is invite-only

Team numbers are public. Finding one proves nothing, so it grants nothing. A member is invited by somebody already on the team — by email, or by a short code that expires and can be revoked.

THE DAILY SCREEN

What is happening, and what is late

Today answers one question — what do I need to do — and it answers it differently for every person who opens it.



Today. Countdown to the next competition, the meeting happening tonight with its RSVP, the tasks assigned to you with subteam and owner, anything waiting on an approval, and the budget headline.

Real progress only

The getting-started checklist ticks itself off when the thing genuinely exists, and the card disappears once the team is running. Nothing to dismiss and re-find.

Late is visible

An overdue task reads as overdue on the day it slips, not at the weekly meeting when it is already a problem.

Approvals are structural

No setting lets a student approve their own purchase. Authority is enforced, not configured.

PLANNING

A planner, not an event list

Build nights repeat. Deadlines do not expect anyone to turn up. A skipped week is not a deletion. The calendar models all three.

The screenshot shows the FTC Home app interface for Team 19645 INFINITY. The user is Aarush Kikkuru, a student. The interface includes a sidebar with navigation options: Today, Calendar (selected), Chat, Build, Live, Season management, Weekly, Roster, Budget & sponsors, Plan, Parts, Scout, Archive, App, How this works, States & sync, and Settings. The main content area displays a calendar for August 2026 with tabs for All, Builds, Comps, Outreach, Deadlines, and Tasks. The calendar shows various events, including Engineering nights, Build nights, and competitions. To the right of the calendar is an 'AGENDA & DEADLINES' section listing events with their dates, times, and locations. The events listed are:

- 17 Aug: Engineering notebook — week 6 write-up (due: Notebook)
- 18 Aug: Build night (18:00 · Room 114 — Tech HS)
- 19 Aug: Rewire odometry pods — strain relief (due: Electrical)
- 20 Aug: Build night (18:00 · Room 114 — Tech HS)
- 21 Aug: Re-cut intake side plates (2mm thinner) (due: Mechanical)
- 22 Aug: Tune AprilTag alignment for far shot (due: Software)
- 23 Aug: STEM night — Lincoln Elementary (17:30 · Lincoln Elementary)
- 24 Aug: Book the van for Naperville (due: Logistics)
- 25 Aug: Build night (18:00 · Room 114 — Tech HS)
- 26 Aug: Practice field reset drill — under 20s (due: Drive team)
- 27 Aug: Build night

Calendar. Meetings that repeat weekly or monthly on chosen days are expanded when read rather than written out as hundreds of rows, so a single occurrence can be skipped without touching the series. Task due dates land on the same grid. The agenda column on the right merges events and deadlines into one ordered list.

Families can subscribe

Export produces a proper .ics feed — folded, escaped, with recurrence rules and exception dates — so a parent's phone calendar shows the season without anybody creating an account.

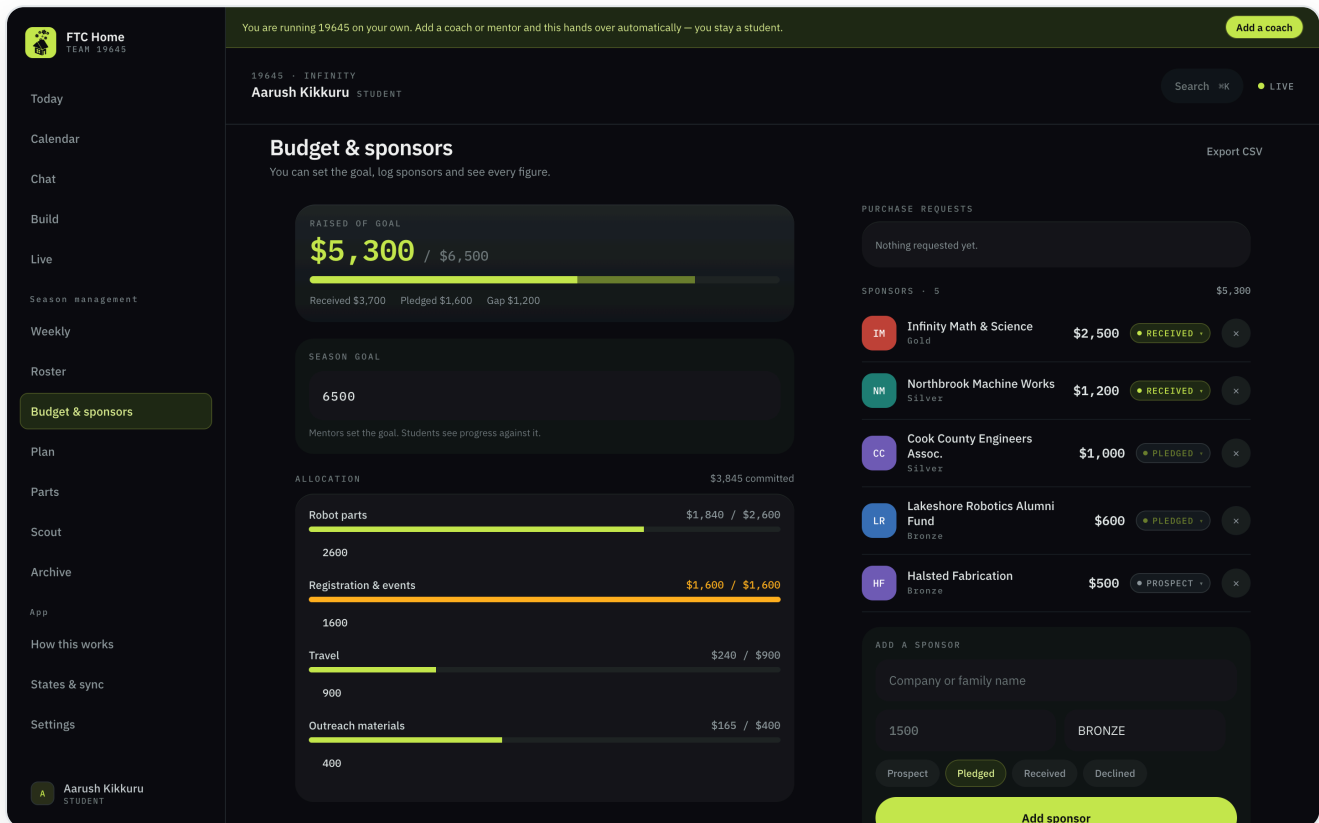
Competitions arrive on their own

Dates and venues come from FTCScout and are cached locally, so the schedule survives a venue with no signal. Locally-created entries are never overwritten by a refresh.

WHERE SPONSOR MONEY GOES

Every dollar, accounted for and visible

This is the screen that matters most to the people funding a team. A sponsor's contribution is tracked from pledge to received to spent, against a goal the team set in public.



Budget & sponsors. Raised against goal, split into received and pledged so an optimistic number never masquerades as money in the account. Allocation caps per category turn amber when a line is fully committed. Sponsors carry a tier and a state. Everything exports to CSV for a treasurer who would rather work in a spreadsheet.

Purchases are requested

A student raises a request, a mentor or a granted approver decides. The trail is kept.

Figures can be narrowed

A coach can restrict budget figures and purchase amounts to signed-in members or to staff.

Nothing is deleted

Finished work moves to an archive, searchable and intact, rather than disappearing.

PEOPLE

Roles that mean something

You are running 19645 on your own. Add a coach or mentor and this hands over automatically — you stay a student. [Add a coach](#)

19645 · INFINITY
Aarush Kikkuru STUDENT

Search 🔍 • LIVE

Roster

8 members · 7 invites pending [Export CSV](#)

• Coach tools on. You can add members, set roles and read contact details.

ADD A MEMBER

Full name

their@email.com

Student Captain Mentor Coach Parent

SUBTEAMS

Mechanical Software Electrical Notebook Outreach Drive team Logistics

+ Add one

Pick as many as apply. Anything you add here shows up for the whole team.

Add member

Puts them on the roster before they have signed in. When they sign in with that address they are matched to this row and go straight in with the role you picked — no second approval. Without an address there is nothing to match them by, so use **Invite somebody** above instead.

NO ADULT HAS SIGNED IN YET

Jordan Calder, Rina Ito are on the roster but have not signed in, so nothing can be approved yet. Send them the site, or an invite code from the roster.

AK	Aarush Kikkuru	CAPTAIN · SOFTWARE · INVITE PENDING	ADD CONTACT	✕
AA	Anish Agrawal	CAPTAIN · MECHANICAL · INVITE PENDING	ADD CONTACT	✕
MR	Maya Rodriguez	STUDENT · ELECTRICAL · INVITE PENDING	ADD CONTACT	✕

Roster. Members, subteams and roles. Contact records and medical notes sit behind a mentor-only gate; a coach can also hand a single capability to a single person by name — so a trusted captain runs the budget without being made a coach.

Adding somebody is an invite

Not an account. They set their own password, or sign in with Google, GitHub or Microsoft, the first time they arrive. Nobody shares a password.

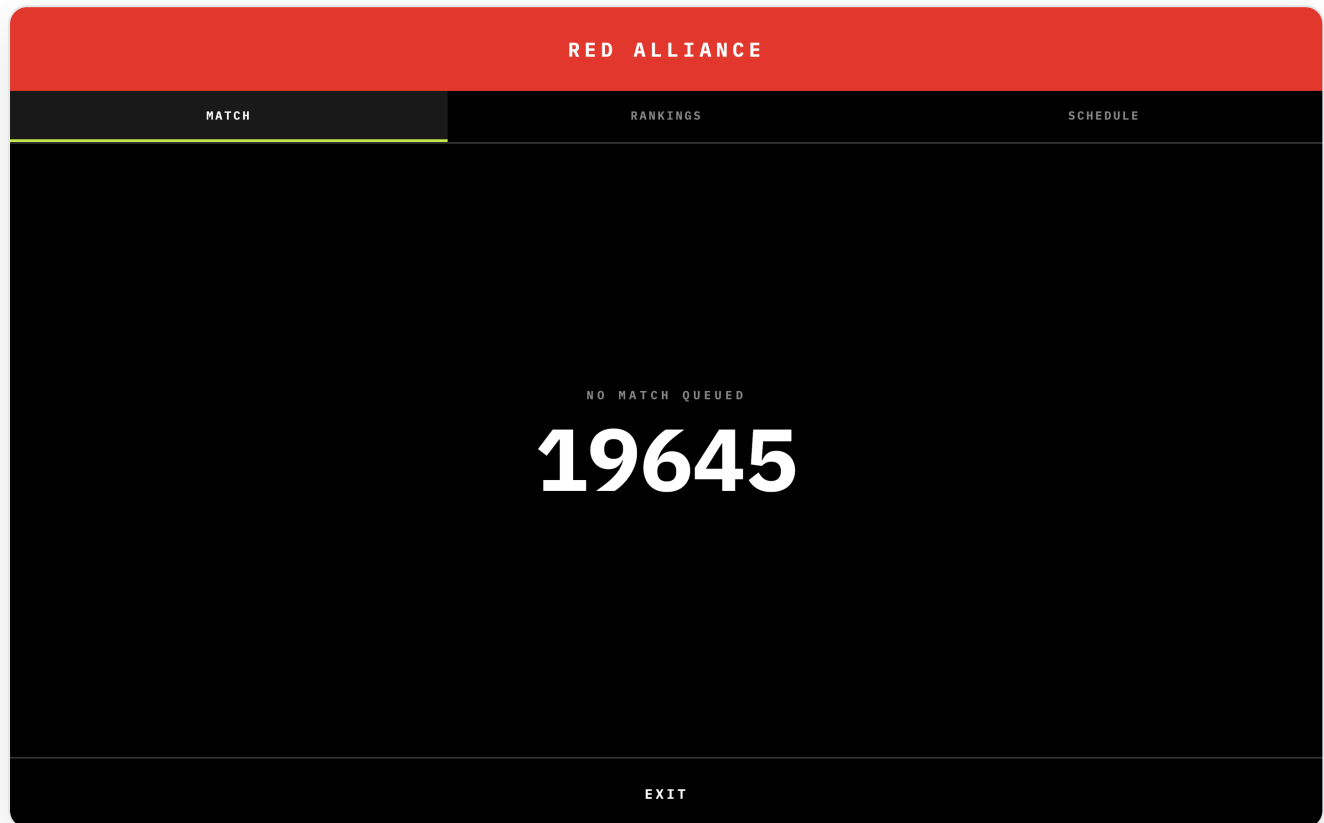
Structural authority is not configurable

No setting lets a student approve their own purchase, and no setting hands a minor's contact details to a signed-out guest. Those are enforced in the database, not by hiding a button.

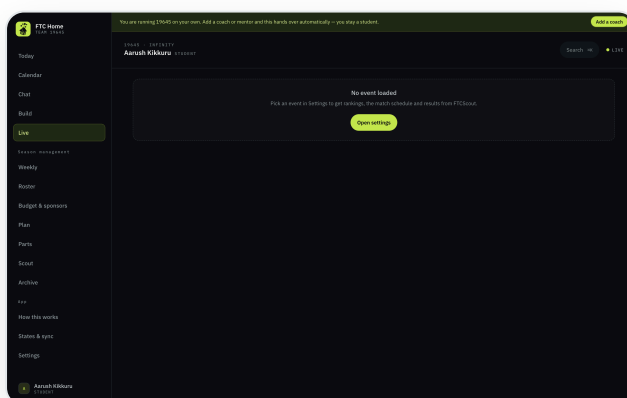
COMPETITION DAY

Built for the worst room in the season

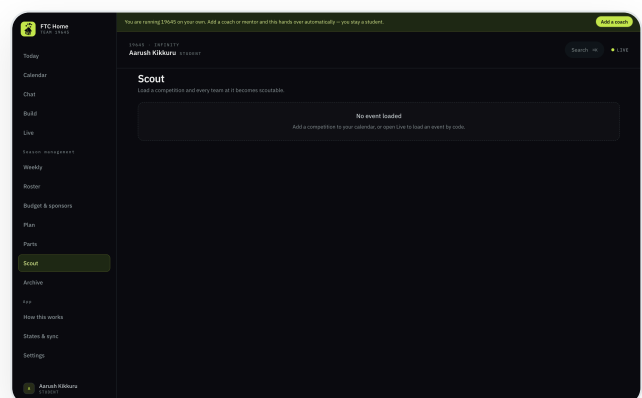
A qualifier is loud, bright, crowded and has no usable wifi. Three screens are designed specifically for it.



Competition Mode. A pit board. Pure black, a match clock at 92px on a phone and 260px on a laptop, alliance colour across the top, and a wake lock so the display does not sleep between matches. One tap from anywhere in the app.



Live. Rank, record, the match queue and scouting cards for whoever you are about to play.

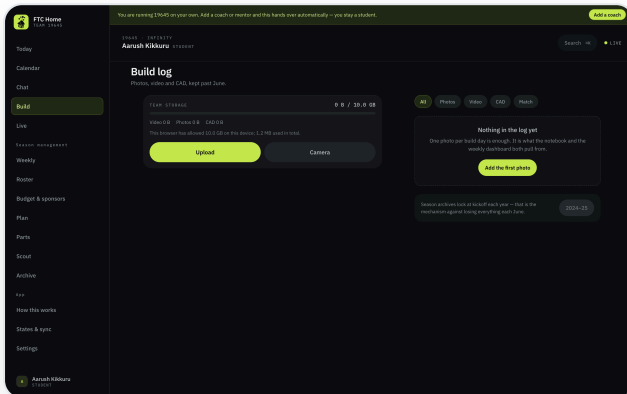


Scout. Every team at the event with rank, record and OPR from FTCScout, plus your own rating and alliance shortlist.

THE RECORD

Evidence, gathered as you go

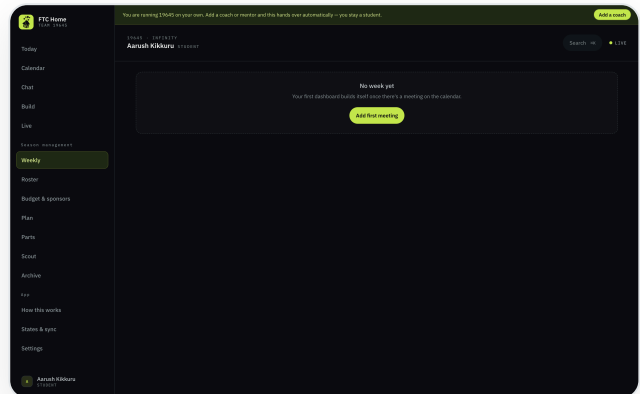
The engineering notebook is judged. Writing it in April from memory is how teams lose awards they earned in November.



Build log. Photo, video and CAD upload straight to the device with generated thumbnails, a storage meter, and an offline queue that drains when signal returns.

A CAD viewer that says what it cannot do

STL and OBJ render in a WebGL viewer written directly against the graphics API and loaded only when needed. Formats that cannot be drawn say so plainly, and say what to export instead, rather than showing an empty canvas.



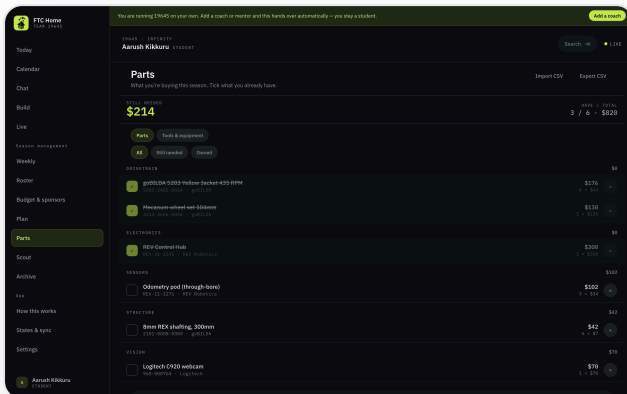
Weekly dashboard. Blocks derived automatically from tasks and attendance, plus blocks written by the captain. Publishes, prints and exports to markdown.

Printed for the people who fund it

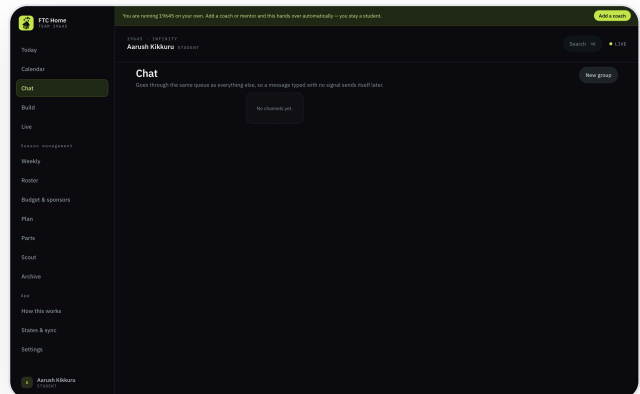
The weekly dashboard is the one thing teams hand to sponsors, so it has a real print stylesheet — it inverts to ink on paper instead of costing a toner cartridge in graphite.

DAY TO DAY

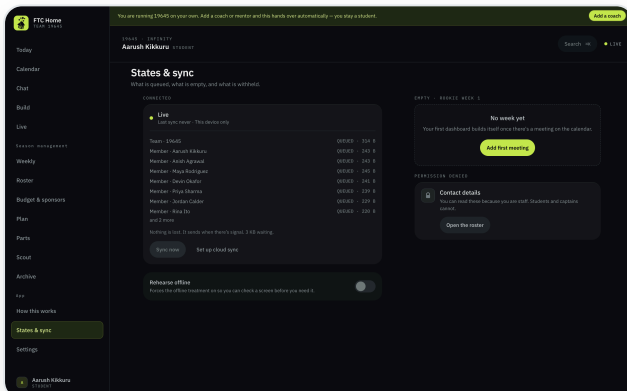
The rest of the season



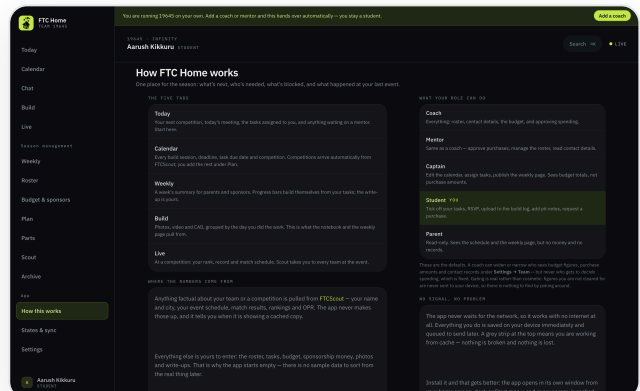
Parts. Your own bill of materials with vendor, part number and price. Imports and exports CSV to the standard, so a vendor sheet with commas in part names survives the round trip.



Chat. Team, subteam and group channels with unread tracking per device, queued offline like everything else.



States & sync. The outbox, in the open: what is queued, how large it is and when it will go. A sync you cannot inspect is a sync you do not trust.

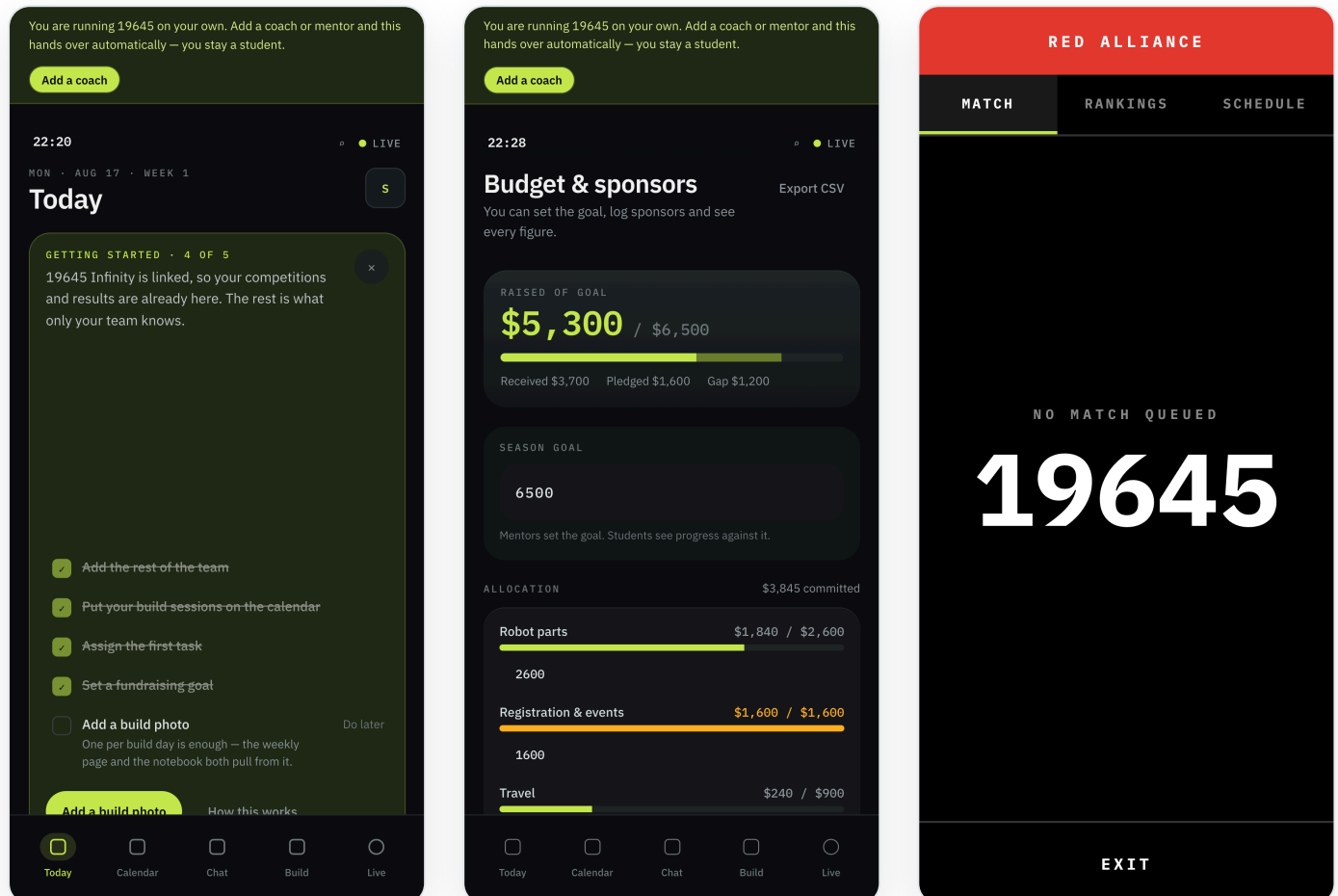


How this works. Where every number comes from and what each role is allowed to do — written for a new member, in the app.

ON A PHONE

Where it is actually used

Students are on phones. The layout is designed at phone width first and expands into a laptop layout — a 240px navigation rail, a three-column Today, a masonry weekly dashboard — rather than the other way around.



It installs

Add to Home Screen on iPhone, or the install button on Chrome and Edge. Its own window, its own icon, no browser bar, no app store.

It opens offline

A service worker caches every screen, and the browser is asked not to evict a season under storage pressure.

It alerts once

Match notifications fire at the lead time, at one minute and at zero, off the real schedule — and never repeat. An alert you learn to swipe away is worse than none.

UNDER THE HOOD

How it is actually built

This is a real engineering artefact, written and reviewed by students. It is the same discipline the robot gets.

19

SCREENS, ALL ON REAL STATE

172

AUTOMATED TESTS PASSING

0

FIXTURES OR DEMO SEASONS

Stack

TypeScript and React, built with Vite. State persists to IndexedDB on the device. Postgres and authentication are provided by Supabase, with access enforced by row-level security policies in the database rather than by checks in the interface.

Offline is the default, not a mode

Reads come from local storage; writes land in an outbox and drain when there is signal. No screen ever awaits the network, so a slow venue never blocks a student mid-match.

Safety decisions we made deliberately

- **Invite-only membership.** A public sign-up page is not also a way onto every team's roster.
- **No shared team password.** It protected nothing that individual accounts and an approval did not already cover.
- **Role escalation is never self-service** — the role that can read minors' contact details cannot be granted to yourself.
- **Minors' records are separated** from the ordinary profile and carry their own restrictive policies.

TYPESCRIPT

REACT

VITE

INDEXEDDB

SERVICE WORKER / PWA

SUPABASE

POSTGRES RLS

WEBGL

WEB NOTIFICATIONS

FTCSCOUT API

CSV / ICALENDAR

GITHUB ACTIONS CI

WHAT YOUR SUPPORT BUILDS

Software is what the season leaves behind

A robot is disassembled in July. What survives is what the students learned building it, and the tools and records they leave for the team that comes next.

FTC Home was written by students on team 19645 to solve a problem we actually had. Along the way it became the kind of work that gets people hired: a real product, with real users, offline-first architecture, a database security model, automated tests and continuous integration. It is also the answer to the question judges ask every year — what did you build that outlasts this season.

What sponsorship funds

Competition registration and event fees, robot parts and materials, travel to qualifiers, and the outreach that brings this programme to younger students in our area. Every category is tracked against a cap in the screens shown in this document.

What a sponsor gets back

Visible placement on the robot, our materials and this software, and a season report generated from real records rather than written at the end. You can see exactly where a contribution went, because we already track it that way for ourselves.

Talk to us. We are happy to demonstrate the app live, walk through the code, or bring the team to you.